Linear Systems

Name _____

Algebra 2 Lesson 1-6 (fast track) Date _____ Period _____

Learning goal: I can solve a system of two linear equations and graph a system of inequalities.
Language goal: I can say "the solution is the ordered pair that makes both equations true" and "the solution region is where the shadings overlap."

On the quiz: You will solve two systems of two equations. Then, for two word problems, you will choose the inequalities that model the situation and pick a point that satisfies all of them.

On the Topic 1 test: Question 12 asks you to solve a system of two equations (Example 1). Question 14 asks you to write and solve a system for bread and egg prices (Example 1). Question 13 (three equations) comes from Example 3, which this packet skips.

Example 1: Solve a system of linear equations

A **system**: two equations, same variables. Its **solution**: the pair (x, y) that makes *both* true.

Solve $\begin{cases} x + 2y = 3 \\ x - 2y = 4 \end{cases}$ **Substitution**

1. Solve each equation for x:
   $x = 3 - 2y$ and $x = 4 + 2y$
2. Both equal x, so set them equal:
   $3 - 2y = 4 + 2y$
3. Solve: $-1 = 4y$, so $y = -\frac{1}{4}$
4. Substitute back: $x = 3 - 2\left(-\frac{1}{4}\right) = \frac{7}{2}$
5. Solution: $\left(\frac{7}{2}, -\frac{1}{4}\right)$

Elimination (same system)

1. The y terms are $+2y$ and $-2y$. Add the equations:
   $(x + 2y) + (x - 2y) = 3 + 4$
2. $2x = 7$, so $x = \frac{7}{2}$
3. Substitute into $x + 2y = 3$:
   $\frac{7}{2} + 2y = 3$, so $2y = -\frac{1}{2}$, $y = -\frac{1}{4}$
4. Same solution: $\left(\frac{7}{2}, -\frac{1}{4}\right)$

Check in both equations:
$\frac{7}{2} + 2\left(-\frac{1}{4}\right) = \frac{7}{2} - \frac{1}{2} = 3$ ✓
$\frac{7}{2} - 2\left(-\frac{1}{4}\right) = \frac{7}{2} + \frac{1}{2} = 4$ ✓

Say it: $(x, y) =$ _____ works in both equations because

Try It 1: Solve each system and check both equations

a. $\begin{cases} 2x + y = -1 \\ 5y - 6x = 7 \end{cases}$

b. $\begin{cases} 3x + 2y = 5 \\ 6x + 4y = 3 \end{cases}$

Example 2: Solve a system of linear inequalities

Malcolm mows lawns for \$20 per hour and walks dogs for \$10 per hour. He wants to earn **at least** \$200 each week, he can work **at most** 20 hours, and he must walk dogs **at least** 5 hours. How many hours of each job meet his goals?

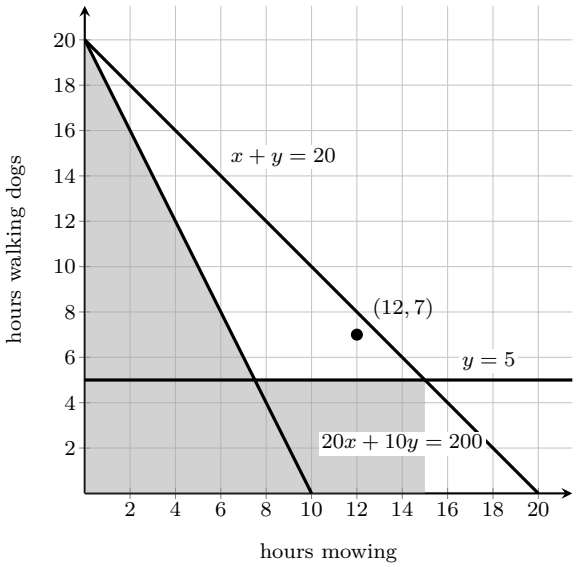
Step 1 Define the variables.
x = hours mowing lawns, y = hours walking dogs
Step 2 Write an inequality for each constraint.

- earn at least \$200: $20x + 10y \geq 200$
- at most 20 hours: $x + y \leq 20$
- at least 5 hours of dogs: $y \geq 5$

Step 3 Graph each boundary line ($\geq$ and $\leq$: **solid**). Test a point to pick the side to shade:

- $20(0) + 10(0) \geq 200$? False: shade *away* from $(0, 0)$.
- $0 + 0 \leq 20$? True: shade toward $(0, 0)$.
- $y \geq 5$: $(8, 8)$ is true, shade above $y = 5$.

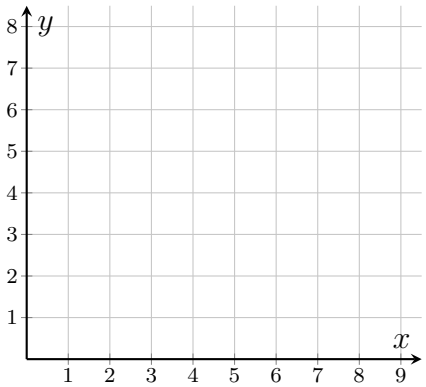
Step 4 The **solution region** is where all the shadings overlap (the shaded triangle).
Check a point. $(12, 7)$ is in the region:
$20(12) + 10(7) = 310 \geq 200$ ✓
$12 + 7 = 19 \leq 20$ ✓ $\quad 7 \geq 5$ ✓
So 12 h mowing and 7 h walking dogs meets every goal.



Try It 2: Graph the solution region

Sketch all points that solve $\begin{cases} 2x + y \leq 14 \\ x + 2y \leq 10 \\ x \geq 0 \\ y \geq 0 \end{cases}$

1. Graph $2x + y = 14$ (solid). Test $(0, 0)$, shade.
2. Graph $x + 2y = 10$ (solid). Test $(0, 0)$, shade.
3. $x \geq 0$, $y \geq 0$: stay in the first quadrant.
4. Darken where every shading overlaps.



Name one ordered pair in your solution region and show it makes all four inequalities true:

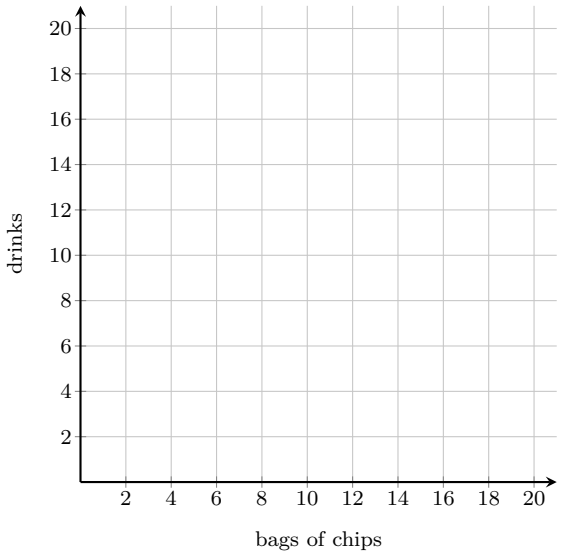
Practice: Model with inequalities

Dmitri is buying snacks for a club meeting. A bag of chips costs \$1.50 and a drink costs \$2.00. He needs **at least 10 items** altogether, and he can spend **at most \$25**. Let x be the number of bags of chips and y be the number of drinks.

a. Write an inequality for each constraint. (Can x or y be negative? Write those too.)
at least 10 items: _____
at most \$25: _____
_____ and _____

b. Graph the boundary lines on the grid. For each line, test $(0, 0)$ and shade the correct side. Darken the solution region.

c. Name one ordered pair in the solution region. Show that it satisfies every inequality, then say what it means (how many chips, how many drinks, total cost).



IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. ⋆ = required. **SmartScore 70 or above on each starred skill replaces your quiz score.**

Code	Skill	Feeds
NJP	Is (x, y) a solution to the system? (warm-up)	warm-up
BW5 ⋆	Solve a system of equations using substitution	quiz 1, 3
2CN ⋆	Solve a system of equations using elimination	quiz 1, 3
U5D ⋆	Solve systems of linear inequalities by graphing	quiz 2, 4
FT6	Solve a system of equations using any method (optional)	quiz 1, 3

Before the quiz: "at least" means $\geq$, "at most" and "up to" mean $\leq$; add $x \geq 0$, $y \geq 0$ when the variables count things; a point is a solution only if it makes *every* inequality true.